

# Statistics 151a: Linear models

UC Berkeley, Spring 2026

Table 1: Calendar (tentative)

| Date   | Day       | Note                   | Unit                           | Topic                                | Assignment   |
|--------|-----------|------------------------|--------------------------------|--------------------------------------|--------------|
| Jan 21 | Wednesday | Lecture                | Introduction                   | Class policies                       | Lab 0        |
| Jan 23 | Friday    | Lecture                |                                | Real-world questions                 |              |
| Jan 26 | Monday    | Lecture                |                                | Inference and probability            |              |
| Jan 28 | Wednesday | Lecture                |                                | Prediction and uncertainty           | Lab 1        |
| Jan 30 | Friday    | Lecture                |                                | Linear algebra review                | HW 0         |
| Feb 2  | Monday    | Lecture                | Unit 1: Multilinear regression | Simple linear regression             |              |
| Feb 4  | Wednesday | Lecture                |                                | Multilinear regression               | Lab (review) |
| Feb 6  | Friday    | Lecture                |                                | ————                                 | Quiz 0       |
| Feb 9  | Monday    | Lecture                |                                | Paths to the least squares objective |              |
| Feb 11 | Wednesday | Lecture                |                                | Transforming the regressors          | Lab 2        |
| Feb 13 | Friday    | Lecture                |                                | Transforming the response            |              |
| Feb 16 | Monday    | Administrative holiday |                                |                                      |              |
| Feb 18 | Wednesday | Lecture                | Unit 2: Testing and inference  | Confidence intervals review          | Lab 3        |
| Feb 20 | Friday    | Lecture                |                                | The multivariate normal              | HW 1         |
| Feb 23 | Monday    | Lecture                |                                | Z and T tests                        |              |
| Feb 25 | Wednesday | Lecture                |                                | Model selection and F-tests          | Lab (review) |

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| Feb 27 | Friday    | Lecture      |                                             | —                                      | Quiz 1       |
| Mar 2  | Monday    | Lecture      |                                             | Asymptotics and heteroskedasticity     |              |
| Mar 4  | Wednesday | Lecture      |                                             | Misspecification                       | Lab 4        |
| Mar 6  | Friday    | Lecture      |                                             | Statistical and practical significance |              |
| Mar 9  | Monday    | Lecture      | Unit 3: Regularization and machine learning | <b>Canceled class</b>                  |              |
| Mar 11 | Wednesday | Lecture      |                                             | Statistical and practical significance | Lab 5        |
| Mar 13 | Friday    | Lecture      |                                             | Splines and basis expansions           | HW 2         |
| Mar 16 | Monday    | Lecture      |                                             | Bayesian estimators                    |              |
| Mar 18 | Wednesday | Lecture      |                                             | Ridge regression                       | Lab (review) |
| Mar 20 | Friday    | Lecture      |                                             | —                                      | Quiz 2       |
| Mar 23 | Monday    | Spring break |                                             | —                                      |              |
| Mar 24 | Tuesday   | Spring break |                                             | —                                      |              |
| Mar 25 | Wednesday | Spring break |                                             | —                                      |              |
| Mar 26 | Thursday  | Spring break |                                             | —                                      |              |
| Mar 27 | Friday    | Spring break |                                             | —                                      |              |
| Mar 30 | Monday    | Lecture      | Unit 4: How regression can go wrong         | Regression to the mean                 |              |
| Apr 1  | Wednesday | Lecture      |                                             | The FWL theorem                        | Lab 6        |
| Apr 3  | Friday    | Lecture      |                                             | Omitted variable bias                  | HW 3         |
| Apr 6  | Monday    | Lecture      |                                             | Outlier responses                      |              |
| Apr 8  | Wednesday | Lecture      |                                             | Outlier regressors                     | Lab (review) |
| Apr 10 | Friday    | Lecture      |                                             | —                                      | Quiz 3       |
| Apr 13 | Monday    | Lecture      | Unit 5: Generalizations                     | <b>Canceled class</b>                  |              |
| Apr 15 | Wednesday | Lecture      |                                             | Final projects                         | Lab 7        |
| Apr 17 | Friday    | Lecture      |                                             | Grouped observations                   | HW 4         |
| Apr 20 | Monday    | Lecture      |                                             | Class review                           |              |

| Date   | Day       | Note    | Unit           | Topic              | Assignment        |
|--------|-----------|---------|----------------|--------------------|-------------------|
| Apr 22 | Wednesday | Lecture |                | Class review       | Lab<br>(review)   |
| Apr 24 | Friday    | Lecture |                | ————               | Quiz 4            |
| Apr 27 | Monday    | Lecture | Final projects | Project discussion |                   |
| Apr 29 | Wednesday | Lecture |                | Project discussion | Lab<br>(projects) |
| May 1  | Friday    | Lecture |                | Project discussion | HW 5              |

Unless otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- [SMT: Statistical Models: Theory and Practice](#) Freedman
- [LME: Linear Models and Extensions](#) Ding
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani
- [VDS: Veridical Data Science](#) Yu, Barter
- [ETM: Econometric Theory and Methods](#) Davidson, MacKinnon

Other books of interest to the class are

- [FPP: Statistics](#) Freedman, Pisani, Purves
- [RDS: R for Data Science](#), Wickham, Grolemund
- [ROS: Regression and other Stories](#) Gelman, Hill, Vehtari